Yanyu Chen

Robotics | Learned Control · Mechanical · Embedded

Contact Information

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GitHub: github.com/yammmyu

Portfolio: https://yanyu-chen.com/

1minute elevator pitch: https://drive.google.com/file/d/18WWm_Fq4reZ2qJb4QDZtAndONWQAj0-e/view?usp=drive_link

Summary

Year 2 NUS Robotics and Machine Intelligence undergraduate. As sole engineer at MUJIN, built and deployed an imitation-learned (diffusion policy) dual-arm manipulation stack on a humanoid robot, covering teleoperation, data collection, real-time control and safety. Part of the Mechanical Team for NUS DJI RoboMaster Calibur Robotics and Captain of the competition division this year. Working proficiency in English (Fluent), Chinese (MotherTone) and German (TELC C1).

Education

National University of Singapore (NUS)
Robotics and Machine Intelligence (B.Eng) | Minor: Innovation and Design
Aug 2025 – May 2029 (expected) | GPA: 4.63/5.00 | Singapore

Frankfurt International School
International Baccalaureate (IB Diploma)
Aug 2019 – May 2025 | Honours: 2023-2025 highest distinction, Physics Award (Class of 2025) |
IB Score: 43/45 (Top 2% Global) | Frankfurt, Germany

Technical Skills

Learned Control & Robotics SW | PyTorch, imitation learning (diffusion policy), YOLO, Ultralytics, OpenCV.
Mechanical Design | SolidWorks, Fusion 360, FEA, DFM/DFA; 5+ years CAD and engineering drawing
Embedded Systems | STM32 HAL/LL, Jetson Orin (CUDA), PID, motor control, CAN/RS485
Robotics Related | ROS2, Gazebo, LiDAR/vision integration
Coding and Tools | Python, C++, CMake, Git, STM32CubeIDE

Experiences

MUJIN Corp.

Software Intern, Humanoid Robotics R&D | May – Jul 2026 | Guangzhou, China
10-week proof of concept: imitation-learned dual-arm parcel handling on an AgiBot Genie G1 humanoid.
Full report, code and engineering documentation: https://github.com/yammmyu/Intern_Summer_2026

- Delivered an autonomous grasp → lift → label-check → flip → place cycle with self-recovery from failed grasps, running end to end from the robot's own three cameras. **~90% task success** at **~12 s median cycle** over 38 scored trials.
- Built from scratch the integration layer between a **10 Hz** diffusion policy and a **120 Hz** arm controller: time-uniform substep expansion, cross-chunk temporal ensembling and Gaussian trajectory smoothing. Replaced clock-based synchronisation with **absolute row IDs** keyed to the robot's own execution clock, eliminating the motion oscillation caused by variable inference latency.

- Designed a layered safety envelope of **11 named invariants** (latched E-stop, per-command displacement watchdog, workspace bounds, mandatory sim validation before any command reaches the robot), enforced by a **pytest suite run as a launch pre-flight** that blocks startup on regression
- Wrote the bilingual (EN/CN) knowhow documentation and PoC report with an explicit verdict: works for the demonstrated task but is not warehouse-ready.

NUS Calibur Robotics (DJI RoboMaster)

Mechanical Team— "Engineer" robot 6-DOF arm | Aug 2025 – Apr 2026

- Designed the Engineer robot's 6-DOF arm, which took **First Prize** at the RoboMaster University League 2026 (Chongqing) Engineer Challenge — 100 points, **2nd place**
- Cut **fasteners by 40%** and **weight by 25%** against the previous generation through truss topology optimization and interlocking parts.
- Designed an internal cable-routing system with conductive slip rings, protecting the wiring and allowing 360° continuous rotation of the end effector.
- Applied design-for-assembly throughout: quick-release structures let any component be replaced in under 3 minutes.

Captain, Competition Team — the team's competition focused division | May 2026 – Present

- Lead the division: set the season phase map, plan projects and assign tasks across members, and own supplier contact and inventory.
- Building a portable moving target, so auto-aim can be tested without tying up a second robot and driver. Planned with a product requirements document and 26-week phased roadmap **(~$910 budget)**

Projects

Ybot - Desktop 6-DOF Robotic Arm

Personal Project | Jun 2023 – Aug 2025 | Mechanical + Firmware

- Started from a market gap: precise arms cost thousands of dollars and hobby kits are imprecise. Planned the whole system, then **designed and built the arm at an estimated S$611 unit cost** (Fusion 360, harmonic drives, 3D-printed PETG) and wrote a C++ ROS 2 node to drive it from MoveIt plans.
- Testing exposed missing joint sensors and belt and axle flaws, which I documented openly; software unfinished.

Bloomcraft — AI bouquet designer (Prototype, not launched)

Personal Project | July 2025 – Aug 2025 | WebApp

- Built after my own failed attempt to order a bouquet: a structured brief returns a **costed arrangement from a priced flower catalogue**, with each flower's cultural meaning attached.
- Utilized React/TypeScript, LLM constrained to JSON, Supabase accounts, **Stripe subscriptions**, in-browser flower recognition, and a Stable Diffusion model fine-tuned over four rounds.

TurtleBot3 Simulated Warehouse Transportation Project

Coursework (CDE2310) | Jan – Apr 2026 | Mechanical + Software

- Owned the **mechanical design** (9-ball gravity-fed magazine and cam launcher within a 5 cm height budget) and **marker-based docking**: pose estimation that resolves the two-solution ambiguity of square markers, plus a one-shot docking plan.

Awards & Certifications

2026 RoboMaster University League (Chongqing)| Engineer Challenge| 2nd Place / 1st Prize

HIMCM Mathematical Modeling Competition | M Award (Top 4 in Europe, 2023)

SolidWorks CSWP (Full Score)

Languages

English: Fluent (academic / professional) **Chinese: Native** **German: Proficient** (TELC C1)